

THE STRUCTURE OF MATHEMATICAL REALITY

Phase I — Chapter 1

Mathematics Is Not About Numbers

Exercises

Self-Study Curriculum

1. Pick two systems from your own experience (biological, data, engineering, anything concrete). Identify a structure they share. Describe the structure without mentioning either system. You have succeeded when someone reading your description cannot tell which systems you were thinking of.

2. Someone says: “Mathematics is just a language.” Construct a two-paragraph argument for why this is wrong, using specific examples. Then construct a one-paragraph argument for why it is *partially* right. Which part is right, and where does the analogy break?

3. Consider the operation of averaging. It is used in biology (mean gene expression), physics (expected energy), economics (GDP per capita), and image processing (blur kernel). These domains have nothing in common materially. What structural property must a system have for averaging to be a meaningful operation on it? When does averaging *fail*, and what does that tell you about the structure of the system where it fails?
